Investigation: BRASIL BROKERS PARTICIPACOES S.A. (CVM code 21180), fiscal year 2018, as it was on 2019-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Emp. Adm. Part. - Const. Civil, Mat. Const. e Decoração (origin: FCA); group: non-financial.

Trading codes (2018 FCA, version 3, filed on 2018-09-13): BBRK3 (Ações Ordinárias, Bolsa, since 2007-10-29).

Document used

2018 DFP, version 1, filed on 2019-03-26; origin: open data; type: consolidated; original scale: MIL.

Versions known on the date: 1 (2019-03-26).

Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=81559

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −2.244 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 74 (n = 355)
- The company's own history: median −3.506; Tukey range (1.5 × IQR): from −5.007 to −1.904; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.288 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 85 (n = 440)
- The company's own history: median 0.964; Tukey range (1.5 × IQR): from 0.527 to 1.274; value above the range (index; 5 fiscal years)
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.100 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 79 (n = 453)
- The company's own history: median 0.997; Tukey range (1.5 × IQR): from 0.963 to 1.041; value above the range (index; 5 fiscal years)
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.994 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 46 (n = 442)
- The company's own history: median 1.033; Tukey range (1.5 × IQR): from 0.869 to 1.203; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

SGI, sales growth index
Value: 1.250 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 78 (n = 455)
The company's own history: median 0.749; Tukey range (1.5 × IQR): from 0.444 to 1.027; value above the range (index; 5 fiscal years)
Confidence in the data: medium

DEPI, depreciation index
Value: 0.917 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 23 (n = 365)
The company's own history: median 0.893; Tukey range (1.5 × IQR): from 0.699 to 1.209; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.722 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 14 (n = 446)
The company's own history: median 1.105; Tukey range (1.5 × IQR): from 0.885 to 1.454; value below the range (index; 5 fiscal years)
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.053 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 43 (n = 469)
The company's own history: median −0.116; Tukey range (1.5 × IQR): from −0.247 to −0.000; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

LVGI, leverage index
Value: 1.282 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 89 (n = 460)
The company's own history: median 0.988; Tukey range (1.5 × IQR): from 0.177 to 2.037; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DPO jump
Value: −96.7 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 30.7 days)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 6 (n = 423)
The company's own history: median −8.6 days; Tukey range (1.5 × IQR): from −136.8 days to 175.8 days; value inside the range (5 fiscal years)
Confidence in the data: medium

Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: zero denominator: average debt (no borrowings in t and t−1)
Alert by the study 1 rule: does not trigger alerts
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: —
Percentile in group and year: —
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: low
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.051 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 46 (n = 466)
The company's own history: median −0.109; Tukey range (1.5 × IQR): from −0.225 to 0.013; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

Financial risk (context only)

Altman Z" (financial risk, context only)
Value: −3.119 (index)
Components: X1 = 0.058; X2 = −1.255; X3 = −0.201; X4 = 1.850 (indices)
Percentile in sector and year: 18 (n = 528); percentile in the non-financial universe: the sector has fewer than 10 companies with a value
The company's own history: median 5.935; Tukey range (1.5 × IQR): from −14.586 to 21.703; value inside the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 3/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): 1
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 1
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

- No long-term debt in either year: the change in leverage (ΔLEVER) is zero and the signal is 0 (project choice; the article only defines fell = 1, rose = 0, p. 8)

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2013, v1: 2014-03-18
- DFP 2014, v1: 2015-03-20
- DFP 2015, v1: 2016-03-29
- DFP 2016, v1: 2017-03-27
- FCA 2018, v1: 2018-01-23
- DFP 2017, v1: 2018-03-28
- FCA 2018, v2: 2018-05-17
- FCA 2018, v3: 2018-09-13
- DFP 2018, v1: 2019-03-26

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
Excerpt checked literally: "Em nossa opinião, essas demonstrações do valor adicionado foram adequadamente elaboradas, em todos os aspectos relevantes, segundo os critérios definidos nesse Pronunciamento Técnico e são consistentes em relação às demonstrações contábeis individuais e consolidadas tomadas em conjunto.Outras informações que acompanham as demonstrações contábeis individuais e consolidadas e o relatório do auditor A Administração da Companhia é responsável por essas outras informações que compreendem o Relatório da Administração."
Auditor in the 2016 FCA: Grant Thornton Auditores Independentes.
Auditor in the 2017 FCA: Grant Thornton Auditores Independentes.
Auditor in the 2018 FCA: Grant Thornton Auditores Independentes.
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2012: Ernst & Young Auditores Independentes S.S. -> KPMG Auditores Independentes. The outgoing firm audited 2 consecutive fiscal years (2010 to 2011), by the start and end dates in the FCA.
- 2016: KPMG Auditores Independentes -> Grant Thornton Auditores Independentes. The outgoing firm audited 4 consecutive fiscal years (2012 to 2015), by the start and end dates in the FCA.

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).

- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:36:41+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.